Homework 4 — Study Notes

Random walks of penetration scores: Law of Large Numbers and Central Limit Theorem

1. Overview

n systems are attacked by m attackers. Each attacker goes through the systems in turn and scores X_i on the i-th one, with X_i i.i.d. The running total $S_k = X_1+\ldots+X_k$ is a random walk, and the m attackers are m independent copies of it. Two models: **Bernoulli** ($X\in\{0,1\}$, $P(X=1)=p$, mean $\mu=p$, variance $\sigma^2=p(1-p)$) and **Rademacher** ($X=\pm1$ with probability ½, $\mu=0$, $\sigma^2=1$, the symmetric case). The task is to compare S_n, S_n/n and $S_n/\sqrt{n}$ as n grows.

2. Theory

Independence gives $E[S_n]=n\mu$ and $\mathrm{Var}[S_n]=n\sigma^2$ (variances add). Scaling by a constant c multiplies the variance by c^2.

Quantity	Mean	Variance	Limit
S_n	$n\mu$	$n\sigma^2$	spread grows like $\sqrt{n}$, no limit
S_n/n	μ	$\sigma^2/n \to 0$	μ in probability (LLN)
$S_n/\sqrt{n}$	$\sqrt{n}\cdot\mu$	σ^2	$N(0,\sigma^2)$ if $\mu=0$ (CLT)

Law of Large Numbers. $S_n/n \to \mu$ in probability: for every $\varepsilon>0$, $P(|S_n/n-\mu|\ge\varepsilon)\to 0$. Proof via Chebyshev: $P(|S_n/n-\mu|\ge\varepsilon)\le \mathrm{Var}(S_n/n)/\varepsilon^2 = \sigma^2/(n\varepsilon^2)$. The bound is valid but loose; the true decay is exponential (Hoeffding/Chernoff).

Central Limit Theorem. $(S_n-n\mu)/(\sigma\sqrt{n}) \to N(0,1)$ in distribution, i.e. $P(\cdot\le z)\to\Phi(z)$ at every z. For Rademacher this is simply $S_n/\sqrt{n}\to N(0,1)$. $\sqrt{n}$ is the only scaling giving a non-degenerate limit: a smaller factor (like 1/n) collapses to a constant, a larger one (like 1) diverges.

Two scales. LLN says where the average sits (μ); CLT says how it fluctuates around it, with size $\sigma/\sqrt{n}$. When $\mu\neq0$, $S_n/\sqrt{n}$ alone is not tight (its mean $\sqrt{n}\cdot\mu$ diverges) and must be centred first.

3. Simulation setup

For each model, m = 5000 trajectories of 10 000 steps were generated; S_n is read at n = 10, 100, 1000, 10000 from the same trajectories. Bernoulli uses p = 0.3, tolerance $\varepsilon = 0.1$. The interactive version is in the web lab (Lab 4).

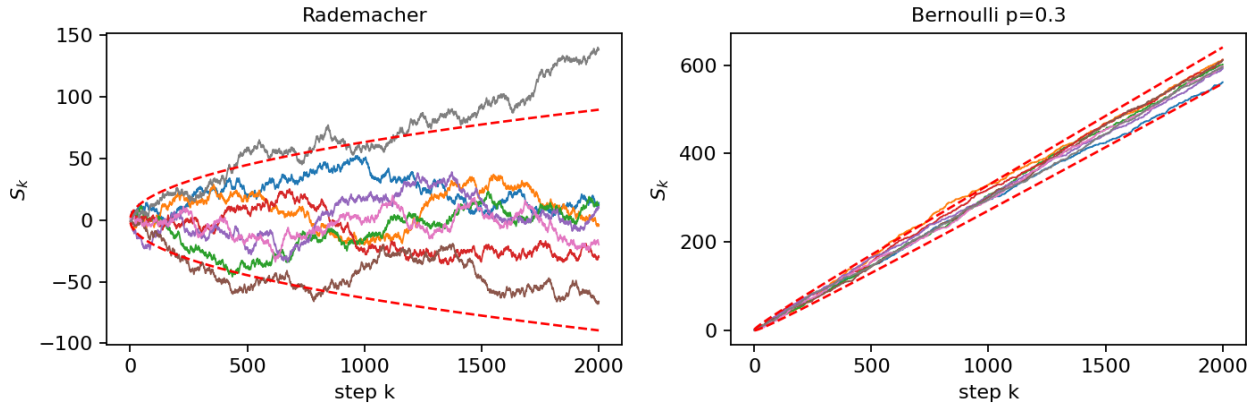


Fig. 1 — Eight trajectories (first 2000 steps) with the envelope $\mu k \pm 2\sigma\sqrt{k}$ (red dashed). Rademacher paths wander around 0; Bernoulli paths follow a line of slope p. The envelope widens like $\sqrt{k}$.

4. Results

Variance (empirical / theory)

Model	n	Var(S_n)	Var(S_n/n)	Var($S_n/\sqrt{n}$)
Rademacher	10	10.2 / 10	0.102 / 0.1	1.02 / 1
Rademacher	100	102 / 100	0.0102 / 0.01	1.02 / 1
Rademacher	1000	1.01e+03 / 1e+03	0.00101 / 0.001	1.01 / 1
Rademacher	10000	9.51e+03 / 1e+04	9.51e-05 / 0.0001	0.951 / 1
Bernoulli p=0.3	10	2.05 / 2.1	0.0205 / 0.021	0.205 / 0.21
Bernoulli p=0.3	100	21.1 / 21	0.00211 / 0.0021	0.211 / 0.21
Bernoulli p=0.3	1000	215 / 210	0.000215 / 0.00021	0.215 / 0.21
Bernoulli p=0.3	10000	2.11e+03 / 2.1e+03	2.11e-05 / 2.1e-05	0.211 / 0.21

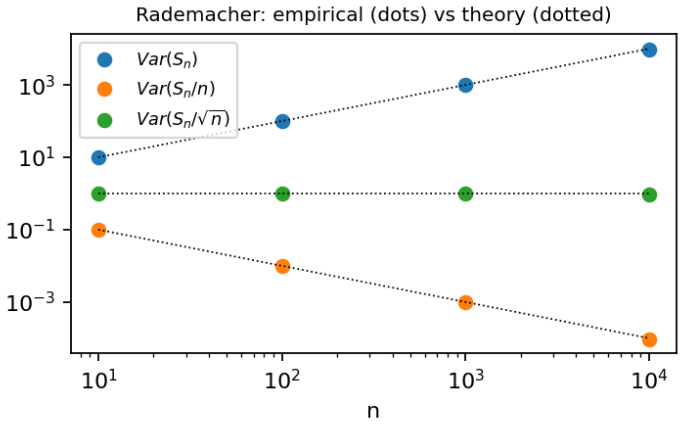


Fig. 2 — Three different slopes on a log-log plot: +1 for S_n, –1 for S_n/n, 0 for $S_n/\sqrt{n}$.

LLN: $P(|S_n/n - \mu| \geq \varepsilon)$

Model	n	mean of S_n/n	empirical	Chebyshev bound
Rademacher	10	-0.00148	0.753	1
Rademacher	100	0.00178	0.378	1
Rademacher	1000	0.000593	0.0012	0.1
Rademacher	10000	8.9e-05	0	0.01
Bernoulli p=0.3	10	0.302	0.501	1
Bernoulli p=0.3	100	0.299	0.0314	0.21
Bernoulli p=0.3	1000	0.3	0	0.021
Bernoulli p=0.3	10000	0.3	0	0.0021

The empirical probability drops to 0 well before the bound does, so Chebyshev is sufficient for the proof but not sharp.

CLT: Kolmogorov–Smirnov distance to N(0,1)

Model	n	mean of $S_n/\sqrt{n}$	KS (standardized)
Rademacher	10	-0.00468	0.125
Rademacher	100	0.0178	0.0545
Rademacher	1000	0.0188	0.0294
Rademacher	10000	0.0089	0.0149
Bernoulli p=0.3	10	0.955	0.146
Bernoulli p=0.3	100	2.99	0.0594
Bernoulli p=0.3	1000	9.48	0.0288

Model	n	mean of $S_n/\sqrt{n}$	KS (standardized)
Bernoulli p=0.3	10000	30	0.0173

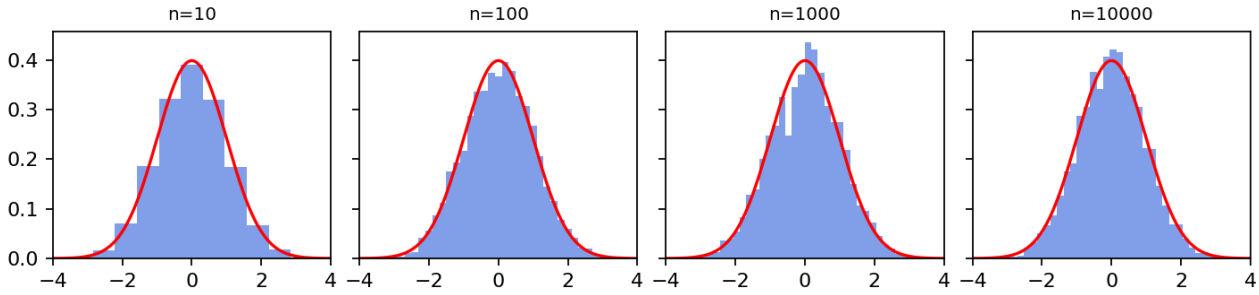


Fig. 3 — Histogram of $S_n/\sqrt{n}$ for Rademacher with the N(0,1) density (red). At n=10 only 11 values exist (discrete); by n≥1000 the fit is close. With m=5000 the KS noise floor is about $1/\sqrt{m} \approx 0.014$ (5% critical value 0.019), so differences below this are not detectable.

5. Interpretation: n systems, m attackers

• **Size n.** The *fraction* of compromised systems S_n/n becomes predictable (variance $p(1–p)/n$), but the *count* S_n becomes less predictable in absolute terms (sd $\sigma\sqrt{n}$). A large infrastructure behaves in aggregate like its average, yet a single attacker's total can still deviate by many systems.

• **Number of attackers m.** Attackers are independent replicas, so m does not alter the distribution of any one S_n; it sets the precision with which we estimate it (errors scale like $1/\sqrt{m}$). Pooling all attackers gives mn trials, with mean $mn\mu$ and variance $mn\sigma^2$, so LLN and CLT apply to the aggregate too.

• **Symmetric case.** With zero mean no attacker has a systematic advantage; about 95% of totals fall within $\pm 2\sqrt{n}$.

• **Assumption.** Independence of systems. Shared credentials or common software introduce correlation, inflating the variance above $n\sigma^2$ and weakening the $\sqrt{n}$ scaling.

6. Exam checklist

• $\mathrm{Var}(S_n)=n\sigma^2$, $\mathrm{Var}(S_n/n)=\sigma^2/n$, $\mathrm{Var}(S_n/\sqrt{n})=\sigma^2$; know why each normalization behaves differently.

• LLN is convergence in probability (Chebyshev proof); CLT is convergence in distribution (not in probability).

• Centre before applying the CLT when $\mu \neq 0$; Bernoulli $S_n/\sqrt{n}$ diverges as $\sqrt{n}\cdot p$.

• Convergence in the CLT is slower for skewed (p far from ½) or discrete small-n cases; KS distance has a floor of $\approx 1/\sqrt{m}$ from finite sampling.

• Rademacher variable: $X=2B–1$ with $B\sim\mathrm{Bernoulli}(½)$, so $S_n^{\mathrm{Rad}} = 2\cdot S_n^{\mathrm{Ber}(½)} – n$.